Project Title:
AcademicAlly

Project Summary:
AcademicAlly is a comprehensive web-based solution aimed at providing insightful feedback to students to improve their academic performances and aid faculties by giving them valuable inferences and pointers to review their teaching methods.

It offers a customized user experience for students, enabling them to submit assignments, showing them an interactive dashboard containing their performance and percentile in each subtopic of a subject and allowing them to provide feedback to teachers. The application also reflects the academic standing of the student relative to their peers and provides additional resources to the student on topics requiring improvement.

The solution also helps teachers grade assignments, send feedback to students and showing them an interactive dashboard containing grade details, class averages and in depth statistics of all students. It also contains an autograder system which grades Multiple Choice Questions instantaneously and provides a plagiarism checker to protect against students copying subjective answers from each other.

Description of AcademicAlly:
This application highlights the granular statistics about student performance in course subtopics which is currently difficult to ascertain and track. It also directs the student towards an improvement path through dedicated recommendations of resources.
The time series plot of the student's relative performance in the subject compared to their peers will motivate them to step up efforts in that subtopic.

The application assists faculties in tuning the level of questions in assignments and final exams by projecting the performance of their class in preceding assignments through an interactive visualization. This will help them maintain a steady difficulty curve throughout the course duration and help students gradually build up their expertise in the subject.

Assesing feedbacks from students is a time comsuming task especially when batch sizes are large. This solution would augment the feedback texts with GenAI powered aggregated feedbacks.

It can be difficult to identify instances of assignment standards being broken, which means that students who are just starting their careers may miss out on opportunities to show integrity. This app provides an informational alert generated via NLP to the faculties for the submissions having high chances of plagiarism. The faculty can match responses of students with similar responses as pointed out by the application prompt.

Creative Component:

- **Dynamic Realtime Analytics Dashboard**
    - The dashboard created will be interactive containing actionable components on click and hover events.
    - The dashboard would visualize the derived metrics from the database data.
- **Plagiarism Checks**
    - Using NLP methodologies, the application would find the text similarities (excluding stopwords) between multiple students answers and notify the faculty about the extent of similarities along with the details of the students between which they're occurring.
- **GenAI**
    - Using NLP methodologies, the application would construct a GenAI powered feedback system, aggregating all the anonymous feedbacks given by students and highlighting the most prominent ones.

Usefullness:

This application provides quantified, action-oriented feedbacks to the students and faculties assisting them to enhance their performance in their respective roles.
Acquiring visibility about the peer performance is useful for managing the effort to be invested for achieving the desireed academic performance.
Student feedback about faculty and courses in a concise aggregation system powered by GenAI significantly reduces the feedback processing time.
Having access to the aggregated student performance metrics is useful for the faculties to tune their assignment difficulty and teaching methods for enabling students to reach desired performances.

Users of the application:-
1) Student
2) Faculty

Basic functions of the application:-

1) Enable Student to perform the following tasks:-
    a) Submit Assignments
    b) ReSubmit/Update Assignments
    c) Submit faculty/course feedback
    d) View Analytics report
    e) View Assignment grades
    f) Subscribe to extra-curricular activities
    g) UnSubscribe from extra-curricular activites

2) Enable Faculty to perform the following tasks:-
    a) Grade Assignments
    b) Create Assignments
    c) Delete Assignments

d) View Analytics report
e) Register to classes
f) View Plagiarism before grading assignment

3) Derive metrics for Students and Faculties:-
   a) Derive Students marks/grade in each subtopic of the subjects.
   b) Detect Plagiarism value by analyzing submitted assignment documents.
   c) Generate aggregated feedback leveraging GenAI from student feedback about faculty/course.
   d) Derive the subtopic for each student in which student has weakest performance relative to peers.
   e) Generate suggested topics and course resources dedicated to the subtopics in which the student is weakest.

Simple features:-

1) Create, Read, Update and Delete operations performed by students and faculties.

Complex features:-

1) Detect Plagiarism value by analyzing submitted assignment documents.
2) Generate GenAI produced aggregated feedback from student feedback about faculty/course.
3) Derive the subtopic for each student in which student has weakest performance relative to peers.
4) Generate suggested topics and course resources dedicated to the subtopics in which the student is weakest.

Existing applications with similar use cases:-

1) Canvas, PraireLearn and Ed

Our USP:-

1) This application provides data oriented insights to all the users of the application which currently is not supported in the existing applications.
2) This application when used, would generate data valuable to research community for evaluating effect of teaching methods on students performance. Our application data can be masked and made open source for research use.

Realness:-

The data sources for the requirements are varied, some being user generated and some being used from the internet which will require modification according to the application's requirements.

The datasets used for the questions in assignments are:

- AI2 Reasoning Challenge (ARC) 2018
    - The ARC dataset is consists of 7,787 science exam questions drawn from a variety of sources which we can leverage to make auto-generated quizzes/assignments which the student can submit and be graded on.
    - The data source is in both CSV and JSONL format. The JSONL files contain the same questions split into the "stem" of the question (the question text) and then the various answer "choices" and their corresponding labels (A, B, C, D). The questionID is also included. The CSV files have the columns - questionID, TotalPossiblePoints, AnswerKey, Diagrams, GradeLevel, Question, Subject and Category.
    - The ARC dataset has a degree of 7787 an a cardinality of 12.
    - **https://allenai.org/data/arc**

- GSM8K
    - The GSM8K dataset consists of 8500 grade school math problems created by human problem writers which we can leverage to make auto-generated quizzes/assignments which the student can submit and be graded on.
    - It has a JSONL structure which consists for question and answer labels.
    - https://github.com/openai/grade-school-math

- University Dataset
    - The dataset contains the information of an imaginary institute with details about its departments, Professors, Courses offered and student performance. This dataset can be modified and used for AcademicAlly to form a basis of the professor details, course details and student details of a school using the application.
    - The degree and cardinality of different CSVs in this dataset together are 5000 and 43.:
    - https://www.kaggle.com/datasets/ananta/student-performance-dataset/data?select=Employee_Information.csv

As these datasets are not tailor made for the uses of AcademicAlly, they will require data augmentation and preprocessing. The application would also use data generated dynamically from its usage.

Detailed description of functionality:-

The detailed functionality of the application from the perspective of users and stakeholders fo the application.

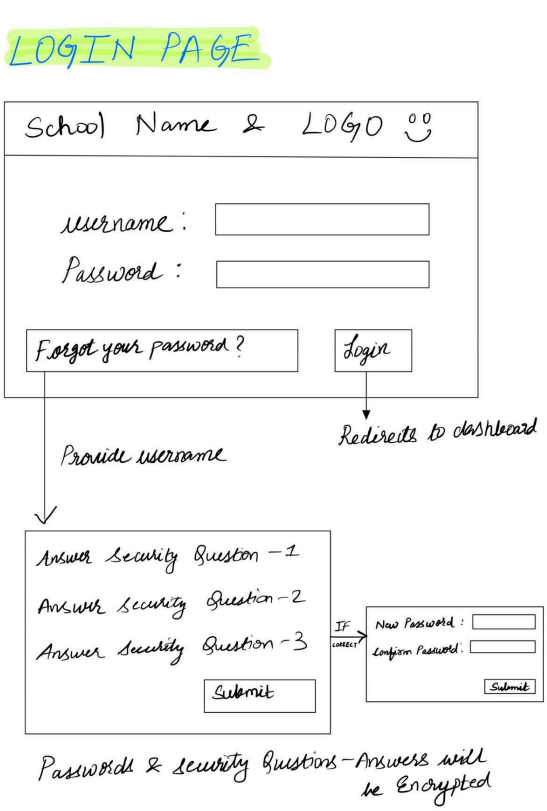
1) Students -
    a) Action1 - Submit Assignments
        i) Students will be able to navigate through the app UI to generate a click event which will be processed by the backend to submit assignments consisting of Multiple Choice Responses and Subjective Answers.
        ii) This will *create* a new object in the Database layer.
    b) Action 2 - ReSubmit/Update Assignments
        i) Students will be able to navigate through the app UI to generate a click event which will be processed by the backend to resubmit or update assignments consisting of Multiple Choice Responses and Subjective Answers until assignment deadlines if enabled by the instructor.
        ii) This will update the existing assignment's object in the database layer.
    c) Action 3 - Submit faculty/course feedback
        i) Students will be submitting text responses from the UI which will be *stored* in the DB as well as queued for an intelligent agent to act on.
    d) Action 4 - View Analytics report
        i) Students will be able to *view* and interact with a personalized dashboard dedicated to visualize their academic performances and application-suggested feedbacks on areas to focus on.
    e) Action 5 - View Assignment grades
        i) Student will be able to *view* their graded assignments after instructor grading.
    f) Action 6 - Subscribe to extra-curricular activities
        i) Students can enroll in extra-curricular activities as listed in the UI. This would *create* a new data object maintaining the number of activities and time-hours of each activity the student is enrolled in.
    g) Action 7 - UnSubscribe from extra-curricular activites
        i) Students can unsubscribe from extra-curricular activities as listed in the UI. This would *delete* the existing data object corresponding to the student.

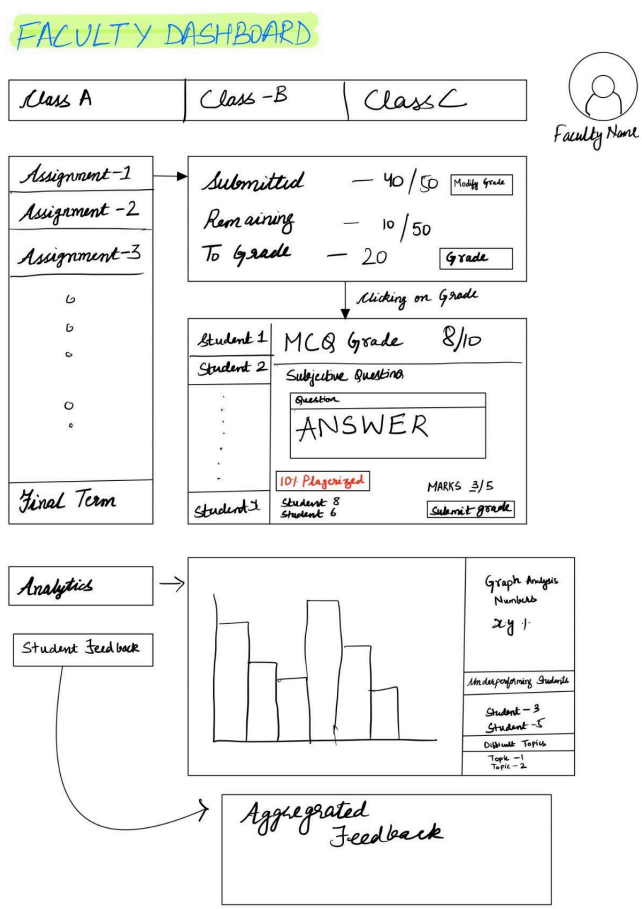
2) Faculties
    a) Action 1 - Grade Assignments
        i) Instructors will be able to navigate through the app UI to generate click events which will be processed by the backend to view and grade submitted assignments.
        ii) While objective answers in the assignment will be autograded by the application, the instructor will have the option to grade the subjective answers in the assignment. The instructor will also be able to view a

plagiarism report generated by the application informing them about similar answers between students and the percentage of similarity.

b) Action 2 - Create Assignments
   i) Instructors will be able to add new assignments via the UI which would create new entries in the DB Table.
c) Action 3 - Delete Assignments
   i) Instructors will be able to delete existing assignments via the UI which would remove those corresponding entries in the database layer.
d) Action 4 - View Analytics report
   i) Instructors will be able to view and interact with a personalized dashboard informing them about class progress and aggregated student feedback.

Low fidelity UI Mockup:-





STUDENT DASHBOARD

Science	Mathematics	English

Assignment -1
Assignment -2
Assignment -3
⋮
Final Term

due date : 07/02/2024

Status : Not Submitted

Start Assignment

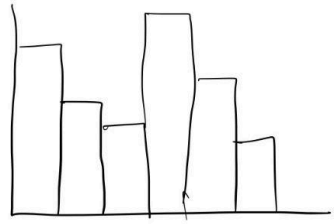
Action Items	
A1	Due Date -1
A2	Due Date -2
A3	Due Date 3
A4	Due Date 4

Question 1	Time Remaining 10:15
Question 2	Subjective Questions
⋮	Question ANSWER
Question x	Submit

Questions can be Multiple Choice or Subjective

Analytics

Course Feedback



Graph Analysis Numbers

Class percentile 82%

Topics to Improve ON

Topic - 1

Topic - 2

Topic - 3

Clicking on the topics redirects to topic resources

Project Work Distribution:-

The application's backend and database layer has a few core functionalities which will be divided between all the members.

Name	Responsbilities
Vaibhav Bhandari	Student Related CRUD functions, Transaction Handling, UI, NLP Services/GenAI, App deployment
Aditya Jhunjhunwala	Student Related CRUD functions, Transaction Handling, UI, NLP Services/GenAI, App deployment
Bhuvanesh Shukla	Faculty Related CRUD functions, Stored Procedures/Triggers, UI, NLP Services/GenAI, App deployment
Gauram Kher	Faculty Related CRUD functions, Stored Procedures/Triggers, UI, NLP Services/GenAI, App deployment